

AWS IAM Role for Same Account Access

To Begin with the Lab

Summary of the Lab

In this lab, **AWS IAM roles** were used to provide temporary access to resources inside the same AWS account. The example focused on a developer named **Amit**, who needed permanent access to **Amazon EC2** for daily work and temporary access to **Amazon S3** when needed. First, Amit was created as an IAM user and given direct EC2 permissions through an AWS managed policy. Then a separate role named **S3 temp access** was created with **Amazon S3 Full Access**, and Amit was added to the trust relationship so he could assume that role only when needed. After switching roles, Amit could access S3 temporarily, and when he switched back, the S3 permissions disappeared again. The lab showed why roles are safer than storing permanent permissions everywhere and how temporary credentials help reduce risk. Pasted text.txt

Understanding IAM Assume Role

An **IAM role** is a temporary identity that can be assumed by a user, service, or application when access is needed for a limited time. It exists so you do not have to give permanent permissions everywhere or keep long-term credentials active all the time. In this lab, Amit used his normal IAM user for EC2 work, but when he needed S3 access, he temporarily switched into the **S3 temp access** role. That temporary session was issued by **AWS STS**, which makes roles safer and easier to manage than assigning full permanent access to every user. Pasted text.txt

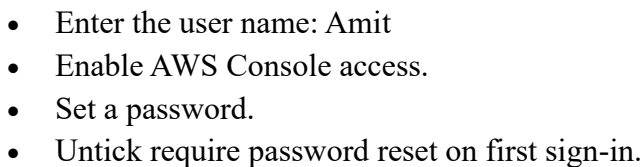
Example:

A developer named Amit works with EC2 every day but only uploads files to S3 occasionally. Instead of attaching S3 permissions directly to Amit's user, the admin creates a role called S3 temp access. Amit switches to that role only when he needs S3, then switches back to his normal EC2 access after the task is complete. Pasted text.txt

Lab Steps

Step 1 – Create the IAM User

- Sign in to the AWS Management Console.
- Open **IAM** → **IAM Users**.



User name
Amit

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and +, =, @, _ (hyphen)

☒ **Provide user access to the AWS Management Console - optional**
Without console access, users with programmatic access and permissions can use the same console credentials for programmatic access without the need for access keys.

Are you providing console access to a person?

User type

☐ Specify a user in Identity Center - Recommended
We recommend that you use Identity Center to provide console access to a person. With Identity Center, you can centrally manage user access to their AWS accounts and cloud applications.

☒ **I want to create an IAM user**
We recommend that you create IAM users only if you need to enable programmatic access through access keys, service-specific credentials for AWS CodeCommit or Amazon Keyspaces, or a backup credential for emergency account access.

Console password

☐ Autogenerated password
You can view the password after you create the user.

☒ **Custom password**
Enter a custom password for the user.
[Masked Password]

• Must be at least 8 characters long
• Must include at least three of the following mix of character types: uppercase letters (A-Z), lowercase letters (a-z), numbers (0-9), and symbols ! @ # \$ % ^ & * () _ + = (hyphen) = [] { } ' "

☐ Show password

☐ **Users must create a new password at next sign-in - Recommended**
Users automatically get the `AWSCloudTrailAssumeRole` policy to allow them to change their own password.

If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user.

- Click **Create user**.

Step 2 – Give Amit Permanent EC2 Access

- Open the **Amit** user.
- Click **Add permissions**.

Identity and Access Management (IAM)

Dashboard

▼ **Access Management**

- Roles
- Policies
- IAM users**
- IAM user groups
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests

▼ **Access reports**

- Access Analyzer
- Resource analysis
- Unused access
- Analyzer settings
- Credential report
- Organization activity
- Service control policies
- Resource control policies

Amit Info

Summary

ARN: arn:aws:iam::[Account ID]:user/Amit

Console access: Enabled without MFA

Access key 1: Create access key

Created: June 26, 2026, 14:50 (UTC+05:30)

Last console sign-in: Never

Permissions | Groups | Tags | Security credentials | Last Accessed

Permissions policies (0)

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type: All types

Policy name | Type | Attached via

No resources to display

Permissions boundary (not set)

Add permissions

- Add permissions
- Create inline policy
- Generate policy based on CloudTrail events

- Choose **Attach policies directly**.
- Search for **AmazonEC2FullAccess** and select the policy.
- Click **Next**.

Step 1
● **Add permissions**
Step 2
○ Review

Add permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

☐ Add user to group
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ Copy permissions
Copy all group memberships, attached managed policies, inline policies, and any existing permissions boundaries from an existing user.

☒ **Attach policies directly**
Attach AWS managed policies directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (1/2642)

Filter by Type: All types 1 match

Search: AmazonEC2Fu

Policy name	Type	Attached entities
AmazonEC2FullAccess	AWS managed	6

Cancel **Next**

- Click **Add permissions**.
- Sign in as Amit.

Provide feedback Multi-session disabled English

aws

IAM user sign in

Account ID or alias (Don't have?)

☐ Remember this account

IAM username

Password

☐ Show Password [Having trouble?](#)

Sign in

[Sign in using root user email](#)

[Create a new AWS account](#)

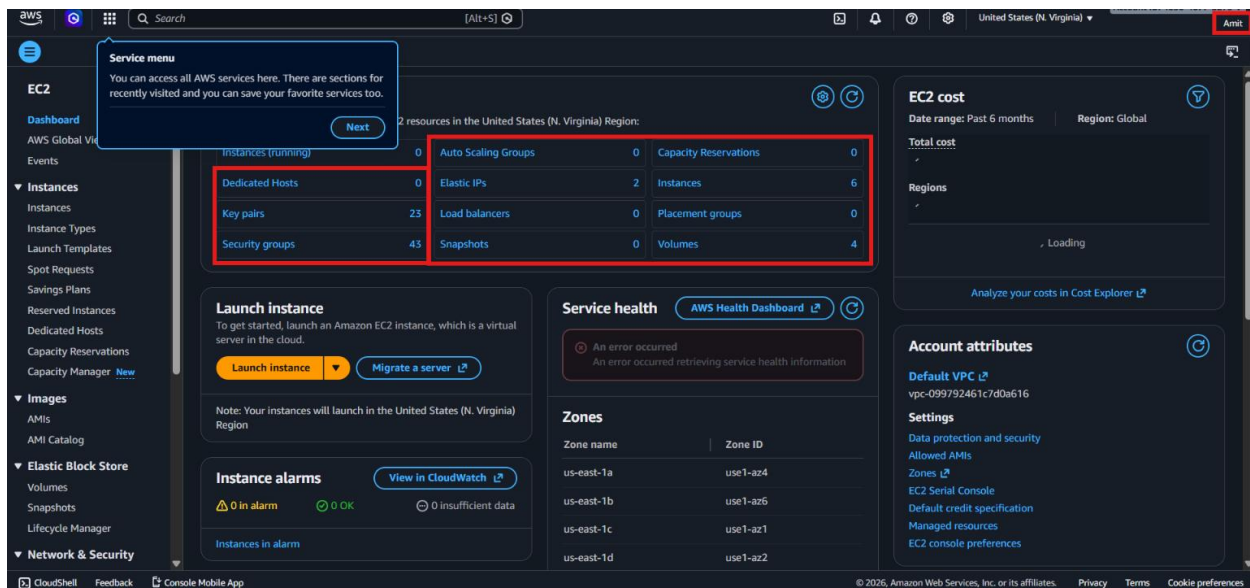
Search foundation for agentic AI

Build agents with real-time retrieval and instant scaling

[Learn more](#)

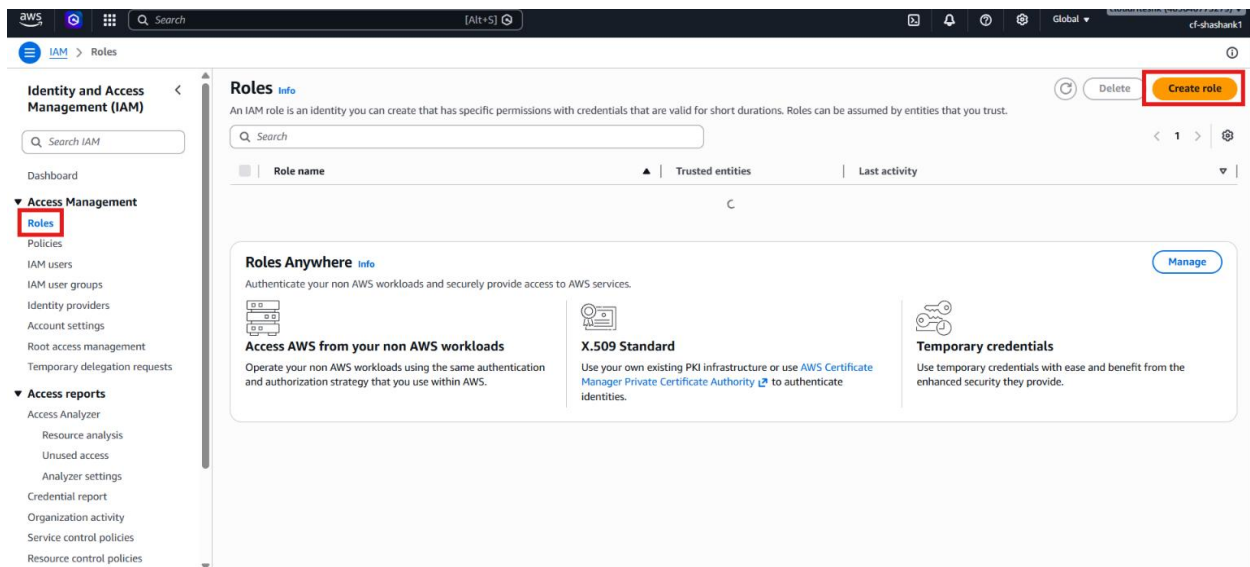
By continuing, you agree to [AWS Customer Agreement](#) or other agreement for AWS services, and the [Privacy Notice](#). This site uses essential cookies. See our [Cookie Notice](#) for more.

- Verify that EC2 resources are accessible.



Step 3 – Create the Temporary S3 Role

- Open IAM → Roles.
- Click Create role.



- Select AWS account.
- Choose This account.

Step 1: Select trusted entity

Step 2: Add permissions

Step 3: Name, review, and create

Select trusted entity

Trusted entity type

- ☐ AWS service: Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☒ AWS account: Allow users from other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- ☐ Web identity: Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ SAML 2.0 federation: Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☐ Custom trust policy: Create a custom trust policy to enable others to perform actions in this account.

An AWS account
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

- ☒ This account (aws-XXXXXXXXXX)
- ☐ Another AWS account

Options

- ☐ Require external ID (Best practice when a third party will assume this role)
- ☐ Require MFA: Requires that the assuming entity use multi-factor authentication.

Cancel Next

- **Attach AmazonS3FullAccess.**

Step 1: Select trusted entity

Step 2: Add permissions

Step 3: Name, review, and create

Add permissions

Permissions policies (1/2299)

Choose one or more policies to attach to your new role.

Filter by Type: All types (1 match)

Policy name	Type	Description
☒ AmazonS3FullAccess	AWS managed	Provides full access to all buckets via the ...

► Set permissions boundary - optional

Cancel Previous Next

- Enter the role name: S3TempAccess
- Create the role.

Step 1 Select trusted entity
Step 2 Add permissions
Step 3 **Name, review, and create**

Name, review, and create

Role details

Role name
Enter a meaningful name to identify this role.
S3TempAccess

Maximum 64 characters. Use alphanumeric and '+', '@', '-', '_' characters.

Description
Add a short explanation for this role.

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: '=', '@', '/', '[', ']', '#', '%', '^', '~', '!'.

Step 1: Select trusted entities [Edit](#)

Trust policy

```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": "sts:AssumeRole",
7       "Principal": {
8         "AWS": "463646775279"
9       },
10      "Condition": {}
11    ]
12  }
13 }
```

Step 4 – Add Amit to the Role Trust Relationship

- Open the **S3TempAccess** role.
- Go to the **Trust relationships** tab.
- Click **Edit trust policy**.

S3TempAccess [Info](#) [Delete](#) [Edit](#)

Summary

Creation date
June 26, 2026, 15:06 (UTC+05:30)

ARN
[arn:aws:iam::111111111111:role/S3TempAccess](#)

Last activity
-

Maximum session duration
1 hour

Link to switch roles in console
[https://console.aws.amazon.com/iam/home?#/roles/S3TempAccess](#)

Permissions **Trust relationships** **Tags** **Last Accessed** **Revoke sessions**

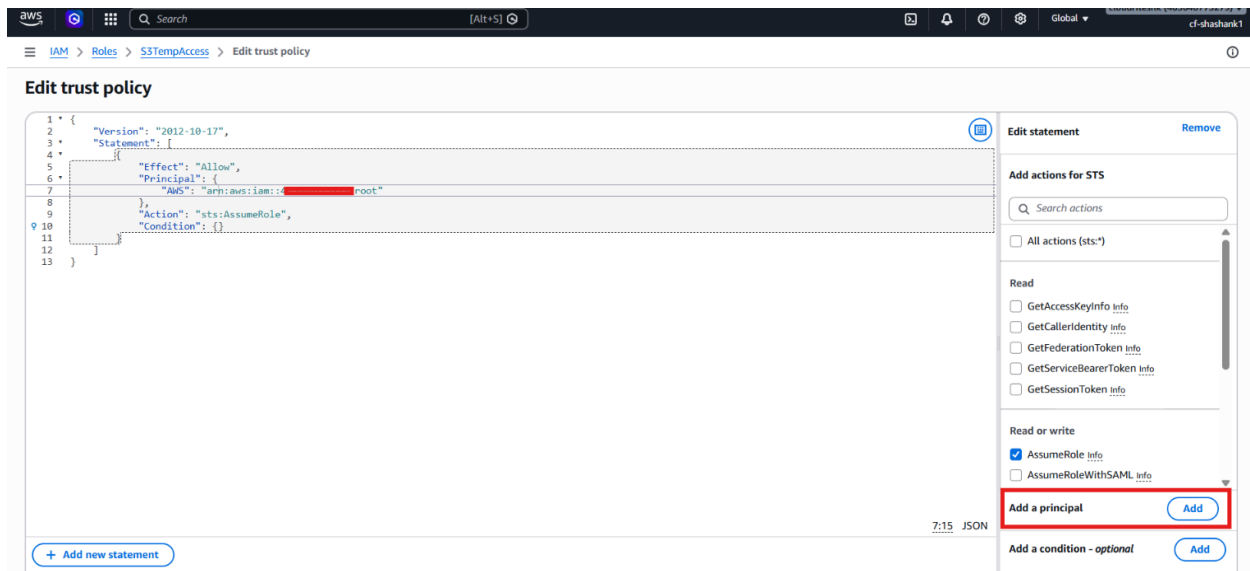
Trusted entities [Edit trust policy](#)

Entities that can assume this role under specified conditions.

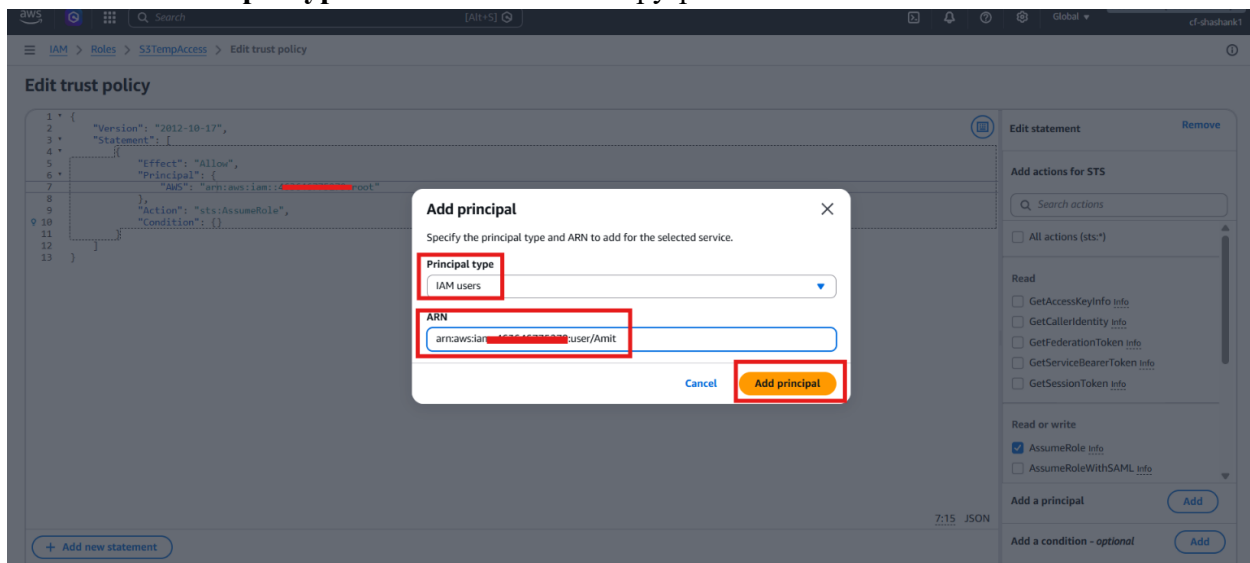
```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": "sts:AssumeRole",
7       "Principal": {
8         "AWS": "arn:aws:iam::111111111111:root"
9       },
10      "Condition": {}
11    ]
12  }
13 }
```

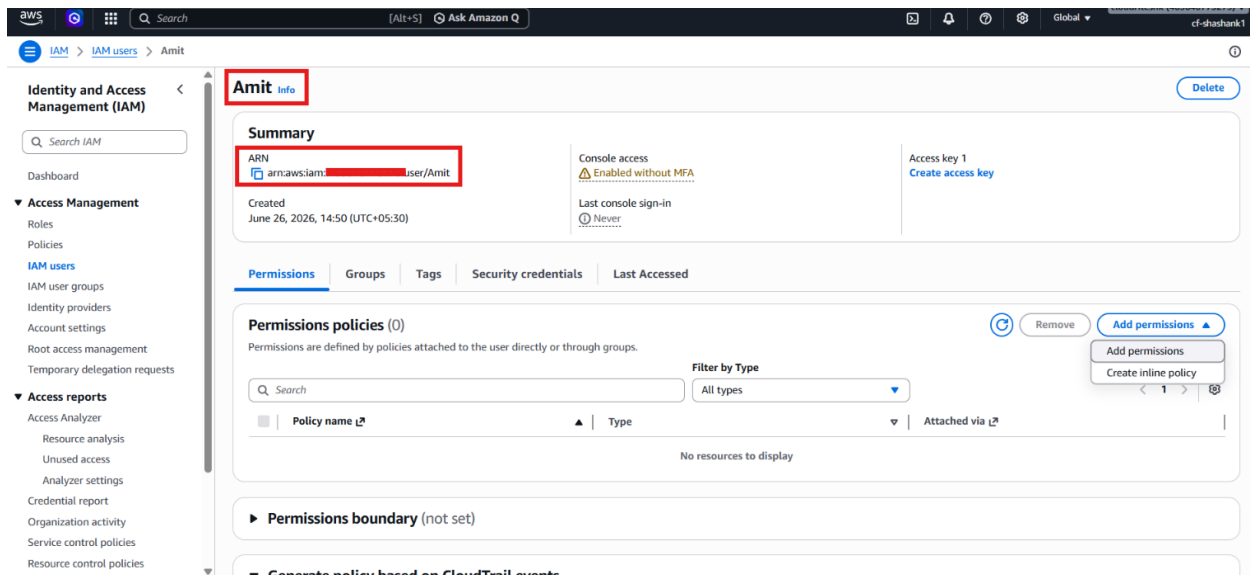
- Add **Amit** as the trusted principal by clicking **Add** on the Add Principal Button.



- Choose **Principal type** as **IAM Users** and copy-paste its **ARN**.



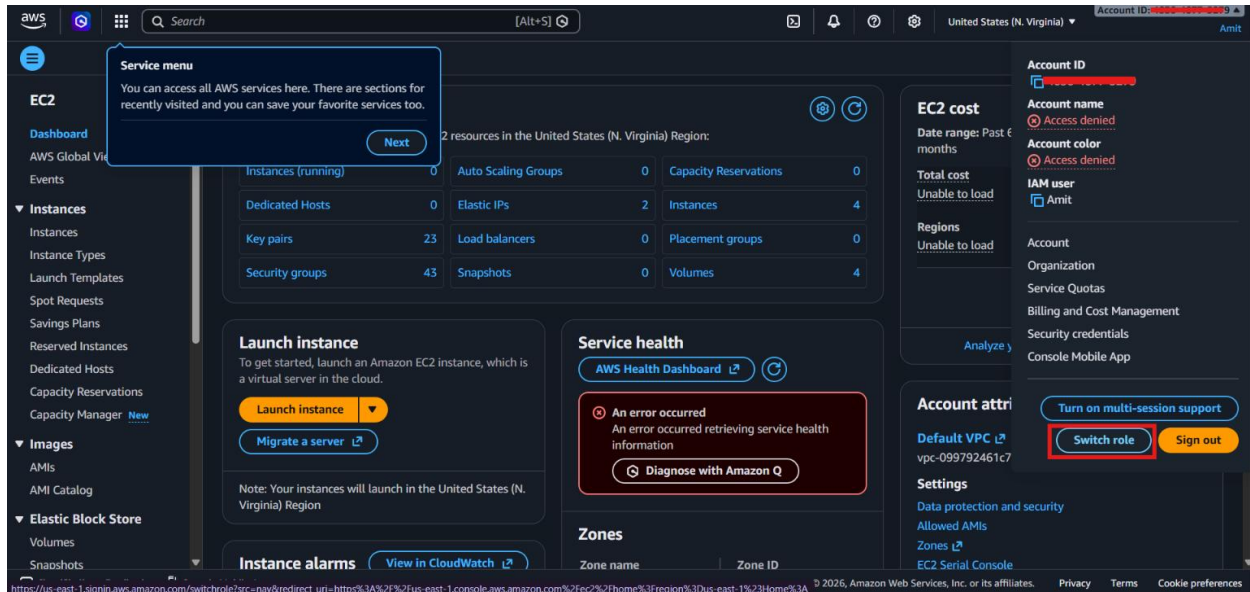
- You can find the **ARN** of **Amit** by opening Amit user details.



- Save the updated trust policy.
- Confirm that only Amit can assume this role.

Step 5 – Assume the Role from Amit’s Console Session

- Sign in as Amit.
- Open the account menu.
- Click **Switch role**.



- Enter the AWS account ID.
- Enter the role name **S3TempAccess**.
- Set a display name if needed.
- Click **Switch role**.

Switch Role

Switching roles enables you to manage resources across Amazon Web Services accounts using a single user. When you switch roles, you temporarily take on the permissions assigned to the new role. When you exit the role, you give up those permissions and get your original permissions back. [Learn more](#)

Account ID
The 12-digit account number or the alias of the account in which the role exists.

IAM role name
The name of the role that you want to assume which can be found at the end of the role's ARN. For example, provide the `TestRole` role name from the following role ARN: `arn:aws:iam:123456789012:role/TestRole`.

Display name - optional
This name will appear in the console navigation bar when active. Choose a name to help identify the permission set assigned to the role.

Display color - optional
The selected color displays in the console navigation when this role is active

☐ None

[Cancel](#) [Switch Role](#)

- Verify that S3 permissions are now available temporarily.

Step 6 – Test Temporary S3 Access

- Open Amazon S3 while assumed into the role.
- Verify that bucket actions are now available.

General purpose buckets (225)

Buckets are containers for data stored in S3.

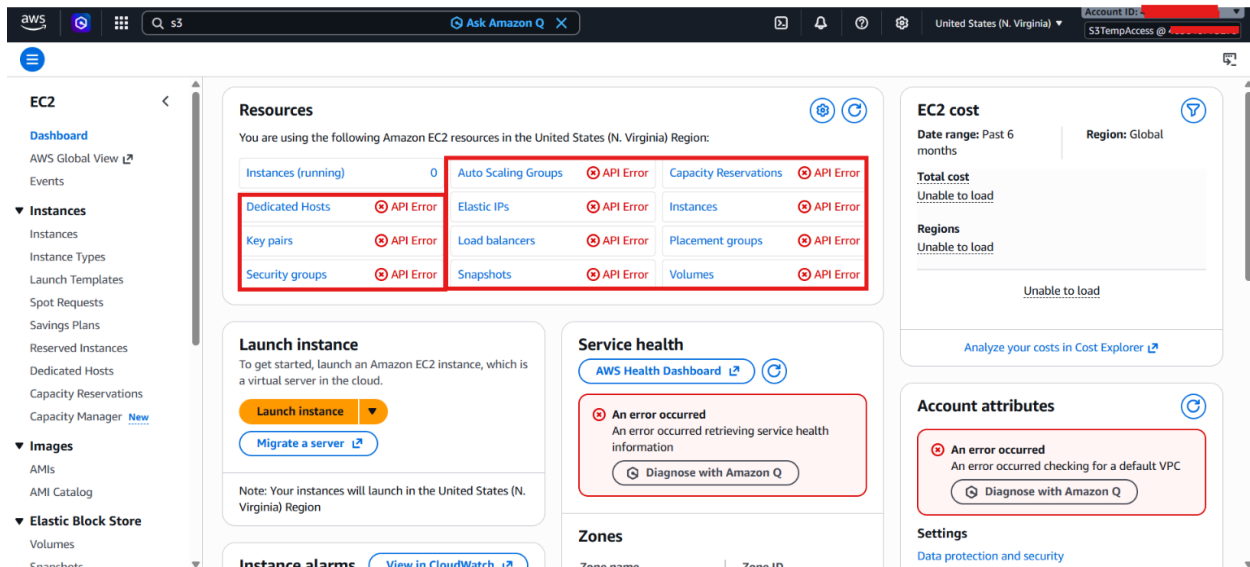
[Find buckets by name](#)

Name	AWS Region	Creation date
arn:aws:s3::test-bucket-1	Asia Pacific (Mumbai) ap-south-1	February 12, 2026, 00:44:57 (UTC+05:30)
arn:aws:s3::test-bucket-2	US East (N. Virginia) us-east-1	June 16, 2026, 15:49:49 (UTC+05:30)
arn:aws:s3::test-bucket-3	Asia Pacific (Mumbai) ap-south-1	June 16, 2026, 15:48:52 (UTC+05:30)
arn:aws:s3::test-bucket-4	Asia Pacific (Mumbai) ap-south-1	November 13, 2025, 15:18:01 (UTC+05:30)
arn:aws:s3::test-bucket-5	Asia Pacific (Mumbai) ap-south-1	January 22, 2026, 11:32:08 (UTC+05:30)
arn:aws:s3::test-bucket-6	Asia Pacific (Mumbai) ap-south-1	November 12, 2025, 11:03:13 (UTC+05:30)

Account snapshot [Info](#)
Updated daily
[View dashboard](#)
Storage Lens provides visibility into storage usage and activity trends.

External access summary [Info](#)
Updated daily
External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

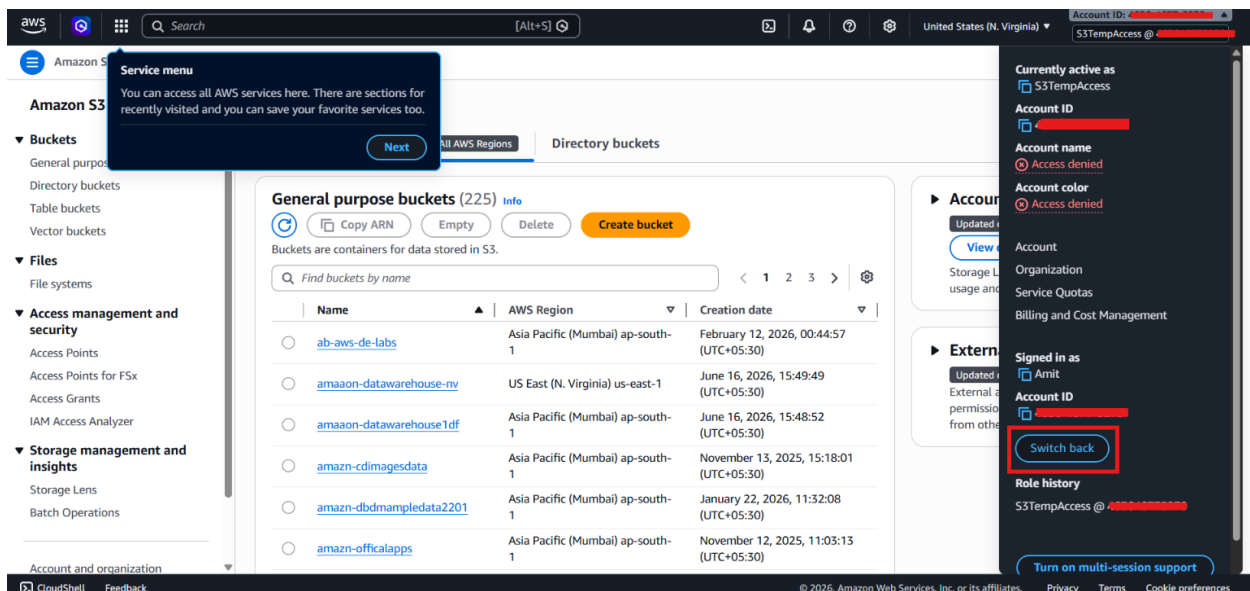
- Create a test bucket if needed.
- Since only S3 is accessible if you now open EC2 it will be unavailable.



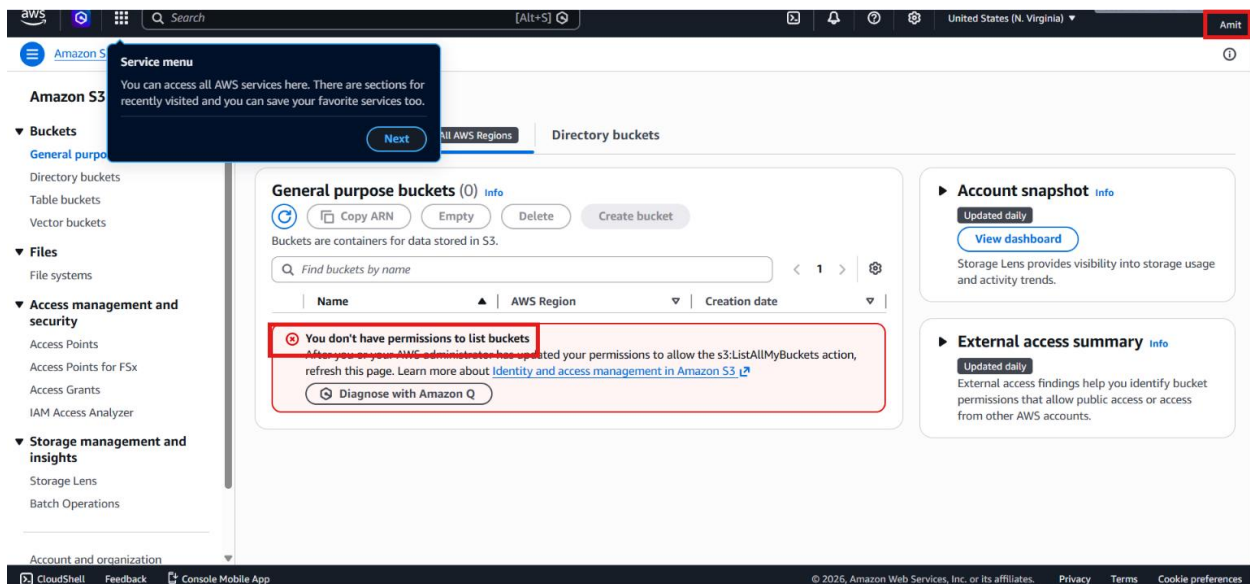
- Confirm that the role gives temporary S3 access only while the session is active for around 1 hour.

Step 7 – Switch Back to the Original IAM User

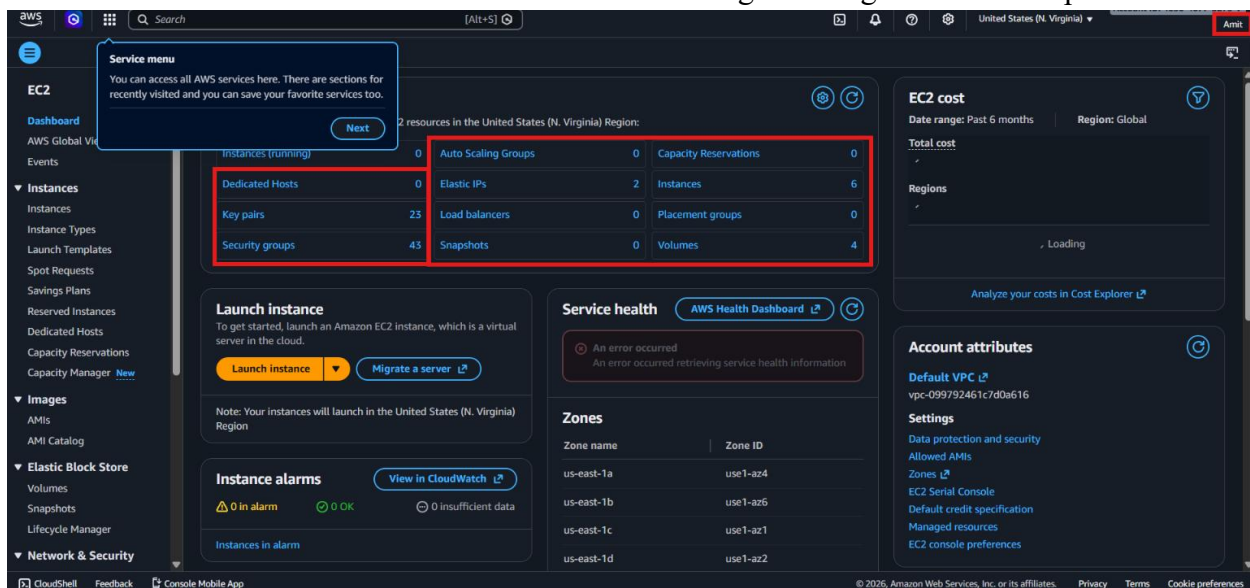
- Open the account menu again.
- Click **Switch back**.



- Return to the normal **Amit** identity.
- Verify that S3 will not open.



- Confirm that EC2 access still remains available through the original IAM user permissions.



Step 8 – Review the Security Benefits

- Understand that role credentials are temporary.
- Review that the default session duration is limited.
- Enable MFA if additional protection is required.
- Use role assumption instead of permanent broad access whenever possible.

Verification

- The IAM user **Amit** was created successfully.
- Amit received permanent EC2 access through an AWS managed policy.
- The **S3 temp access** role was created successfully.
- Amit was added to the role trust relationship.

- Amit could assume the S3 role temporarily.
- S3 permissions were available only while the role was active.
- Switching back removed the temporary S3 access.
- The lab demonstrated same-account IAM role assumption correctly.